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Pareto-optimal estimation and policy learning for balancing short-term and long-term outcomes

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ABSTRACT

In high-stakes domains such as precision medicine and personalized recommendation, optimizing treatments or policies demands a careful balance between immediate rewards and long-term outcomes. However, these objectives often exhibit inherent tradeoffs — for instance, aggressive medication dosages may accelerate short-term recovery yet induce severe long-term side effects. Existing methods struggle to address such dilemmas due to two primary challenges: the lack of explicit mechanisms to reconcile conflicting outcomes and the gradient interference inherent in joint optimization. To address these issues, we propose a Pareto-efficient framework comprising Pareto-Optimal Estimation (POE) and Pareto-Optimal Policy Learning (POPL). Specifically, POE employs a continuous Pareto optimization module to explicitly resolve task-level conflicts between representation learning and multi-outcome prediction. Building upon these counterfactual estimates, POPL explores the treatment-response surfaces over continuous dosage spaces and identifies the Pareto frontier, thus enabling proactive and balanced decision-making. Extensive experiments on both synthetic benchmarks and real-world datasets demonstrate that our approach consistently achieves a superior performance in terms of counterfactual prediction and policy learning.

1. Introduction

In causal inference, accurately estimating treatment effects on target outcomes is foundational for interpretable and reliable downstream decision-making tasks. While traditional research has focused on isolated outcomes, many real-world scenarios — such as healthcare, education, marketing, and social science (Chetty et al., 2011; Hu et al., 2023; Yang et al., 2024) — require the simultaneous evaluation of short-term rewards and long-term consequences. Such short-term effects are readily observable, whereas long-term outcomes remain difficult to study due to prohibitive costs and extended observation periods. Moreover, the short- and long-term outcomes can conflict with each other. As illustrated in Fig. 1, consider antidepressant dosage decisions: clinicians may prioritize short-term metrics like symptom relief and functional im-

provement within the initial months. However, higher dosage, while hastening early recovery, can trigger severe delayed side effects such as drug resistance or lasting health complications after years. Therefore, optimizing treatments is not merely a matter of bias reduction, but a complex issue of multi-objective trade-off.

Recent advancements have introduced various frameworks to estimate long-term effects, typically by leveraging random control trials (RCTs) via data fusion. A key thread employs short-term outcomes as surrogates (Athey et al., 2020, 2025b; Chen & Ritzwoller, 2023; Kallus & Mao, 2022) for inferring long-term impacts under specific independence assumptions, which is illustrated in Fig. 1(a). However, these approaches neglect direct treatment influences on long-term outcomes — a scenario prevalent in practice, as depicted in Fig. 1(b,c). While recent studies (Ghassami et al., 2022; Hu et al., 2023; Imbens et al., 2023) have

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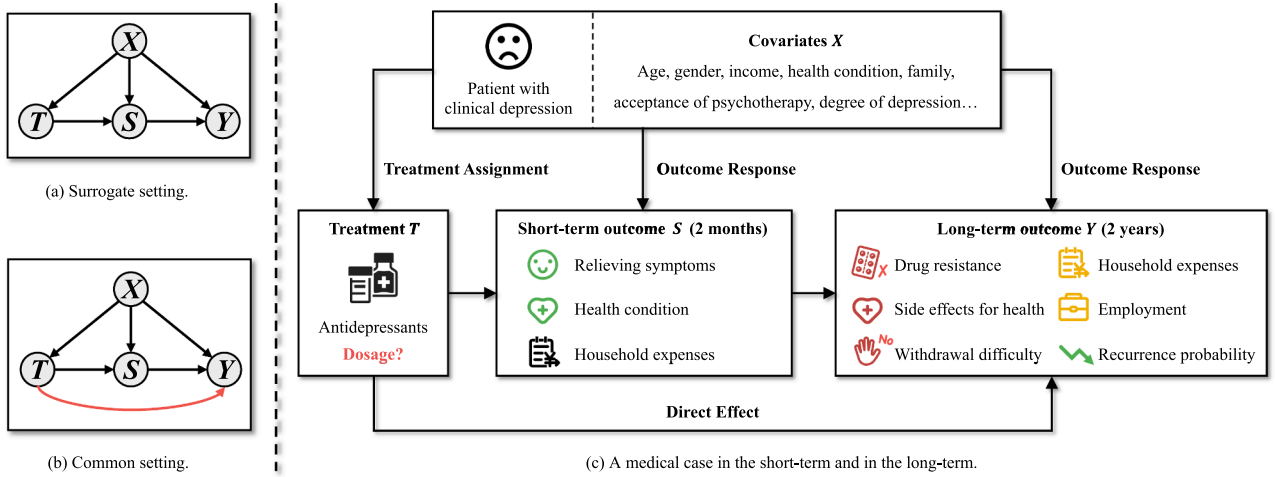


Fig. 1. Illustration of settings in the long-term treatment effect estimation. (a) In the surrogate setting, the short-term outcome S serves as a mediator to block the total effect of treatment T on the long-term outcome Y . (b) In the common setting, the direct influence from T to Y is considered and highlighted in red. (c) We give a medical case to illustrate the importance of trade-off between the short-term and long-term outcomes that may conflict with each other.

relaxed the surrogacy assumption, they still rely on RCT data that may suffer from prohibitive costs, ethical constraints, or limited external validity.

However, all these methods rarely tackle the conflicts between short- and long-term objectives. On one hand, treatments may enhance short-term rewards while simultaneously impairing long-term benefits, creating irreducible trade-offs among multiple outcomes. A real-life example is the administration of medications as provided in Fig. 1. On the other hand, joint training for conflicting tasks (e.g., representation learning and prediction) often suffers from gradient interference, undermining overall optimization. Furthermore, most existing studies emphasize binary treatments with separate outcome regressions, which can result in substantial information loss (Kohavi et al., 2012). Consequently, effectively reconciling short- and long-term causal effects to derive optimal treatments continues to pose an open problem.

To bridge these gaps, we propose a Pareto-efficient framework¹ to determine a treatment dosage whose short- and long-term outcomes are Pareto-optimal. This framework consists of two components: Pareto-Optimal Estimation (POE) and Pareto-Optimal Policy Learning (POPL). Specifically, POE incorporates a continuous Pareto optimization module to directly mitigate task-level conflicts between shared representation learning and multi-outcome predictions. Based on the resulting counterfactual estimates, POPL then maps the treatment-response surfaces across continuous dosage spaces and explores the Pareto frontier. In this way, our proposed method is able to support proactive and balanced decision-making.

In summary, our contributions include:

- We address the under-explored challenge of jointly handling estimation and policy learning under inherent conflicts between short- and long-term treatment effects. Our framework extends beyond conventional surrogacy-based or single-objective methodologies and obviates the need for costly RCT data.
- We develop a Pareto-efficient algorithm featuring POE and POPL modules. It employs Pareto optimization to resolve task-level conflicts and discover optimal treatment-response frontiers.
- We empirically evaluate our proposed framework on synthetic benchmarks and real-world datasets. It consistently demonstrates improved performance in both counterfactual prediction and policy learning.

2. Related work

The rapid advancement of neural networks has significantly strengthened causal inference techniques, as evidenced by invariant representations learned via multi-layer perceptrons in CFR (Shalit et al., 2017), generative adversarial networks in GANITE (Yoon et al., 2018), variational auto-encoders for latent structure discovery in CE-VAE (Louizos et al., 2017), and reinforcement learning for causal structure identification (Amirinezhad et al., 2022). Conversely, causal inference has enriched neural network applications in domains such as healthcare (Li et al., 2024; Zheng et al., 2024), out-of-distribution generalization (Jia et al., 2024; Wang et al., 2024; Yang et al., 2023), and multi-outcome policy learning (Davoudi et al., 2021; Fu et al., 2024; Han et al., 2023; Hervella et al., 2024; Tian et al., 2023; Zhao et al., 2023). These intersections underscore the dynamic evolution at the interface of data science and causal reasoning. In what follows, we review three lines of research most relevant to our work.

2.1. Long-term treatment effect estimation

Data fusion is the most established paradigm where short-term outcomes from RCTs are combined with long-term outcomes from observational data to mutually enhance estimation. Two assumptions are usually discussed, including *surrogacy* and *unconfoundedness*. Surrogacy-based approaches (Athey et al., 2025b; Cai et al., 2024; Cheng et al., 2021; Huang et al., 2023; Hung & Chen, 2026; Lal et al., 2026; Zeng et al., 2024) regard short-term outcomes as surrogates (or mediators) for long-term outcomes, assuming that the long-term outcome is independent of treatment conditional on the surrogates. It is illustrated as Fig. 1(a). However, validating short-term surrogates that fully mediate the treatment effect remains a long-standing challenge (Chen et al., 2007; Frangakis & Rubin, 2002; Ju & Geng, 2010; Lauritzen et al., 2004; Prentice, 1989). Moreover, the well-known surrogate paradox can arise, where the treatment exerts positive effects on both the short-term surrogate and the long-term outcome individually, yet produces an overall adverse effect on the outcome of interest (Chen et al., 2007).

Kallus and Mao (2022) have sought to relax *surrogacy* assumption and explicitly model direct treatment influences on long-term outcomes, corresponding to scenarios in Fig. 1(b). This line has been extended by further relaxing *unconfoundedness* on observational data (Athey et al., 2020; Chen & Ritzwoller, 2023; Hu et al., 2023). However, these methods assume that unobserved confounders do not directly affect long-term outcomes. Consequently, the problem is reduced to addressing a

¹ Codes are available at <https://github.com/suikateiou/POE-POPL>.

Missing At Random (MAR) issue via data fusion, where external validity is ensured through the *comparability* assumption. Further relaxation is achieved through other milder assumptions (Athey et al., 2025a; Ghassemi et al., 2022) or latent representation learning (Cai et al., 2025).

Continuous treatment settings extend the problem beyond binary interventions to dosage-level effects. Recent works also rely on assumptions like MAR (Singh, 2022), *surrogacy* (Goffrier et al., 2023; Zeng et al., 2024), and *latent unconfoundedness* (Yang et al., 2024b). A more general time-series extensions (Chen et al., 2025) consider short- and long-term outcomes as evolving over multiple time steps rather than two fixed horizons. However, it also rely on assumption similar to *Equi-Confounding Bias*.

Overall, existing methods rarely model or resolve inherent conflicts between short-term and long-term outcomes. They also provide limited support for downstream decision-making tasks that proactively identify balanced policies. In contrast, our work utilize observational data under continuous dosage spaces, and explicitly address outcome conflicts through a Pareto-efficient framework. It mitigates task-level conflicts and gradient interference during joint training among representation learning and multi-outcome predictions, and finally traces the Pareto frontier to enable optimal policies.

2.2. Multi-task learning

Estimating short- and long-term treatment effects can naturally be framed as a multi-task learning problem with conflicting objectives. Sener and Koltun (2018) introduced the Multi-Gradient Descent Algorithm (MGDA), which reformulates multi-task learning as multi-objective optimization and seeks Pareto-optimal solutions. However, MGDA typically returns only a single solution, limiting its practicality. To obtain a diverse set of trade-offs, Lin et al. (2019) proposed Pareto MTL, which decomposes the problem into multiple constrained sub-problems corresponding to different preference weights. Nevertheless, this approach requires separate training for each solution and does not fully exploit the continuity of the Pareto frontier. Ma et al. (2020) addressed this by progressively exploring additional solutions from an initial Pareto point, though at the cost of high computational complexity. XWC-MGDA (Momma et al., 2022) improves efficiency by enabling flexible exploration of the Pareto frontier from arbitrary reference points. A recent comprehensive survey on multi-task learning (Zhang & Yang, 2022) further summarizes theoretical advances and future directions in this area.

However, these methods are primarily designed for general prediction tasks and do not address critical challenges in causal inference, such as confounding bias and counterfactual prediction. In this paper, we integrate Pareto principles into short- and long-term causal inference, enabling task-level trade-offs in a gradient-friendly manner. Moreover, it can be naturally extended to scenarios with multiple outcomes or time-series data.

2.3. Policy learning

In this work, we focus on determining the Pareto-optimal policy to trade-off between short-term and long-term outcomes. The purpose is to maximize their total reward. Studies with observational data can be broadly categorized into two main streams: statistics (Luedtke & Van Der Laan, 2016; Qian & Murphy, 2011; Zhang et al., 2012) and machine learning (Beygelzimer & Langford, 2009; Kallus, 2018; Swaminathan & Joachims, 2015). The first category primarily tackles empirical maximization problems and delves into their various relaxations, aiming to refine the theoretical underpinnings of policy learning. The second category concentrates on enhancing the practical performance of policy learning, with a particular emphasis on the application of doubly robust objectives. Wu et al. also consider the policy learning for balancing short-term and long-term rewards (Wu et al., 2024; Yang et al., 2024a)

Table 1

Key variables and notations used in this work.

Notation	Description
$n \in \mathbb{Z}^+$	number of units in dataset $\mathbb{D}$
$i \in \{1, \dots, n\}$	indicator of n units
$X \in \mathcal{X} \subset \mathbb{R}^{d_X}$	covariates of d_X dimensions
$T \in \mathcal{T} \subset \mathbb{R}$	continuous treatment
$S \in \mathcal{Y} \subset \mathbb{R}$	short-term outcome
$Y \in \mathcal{Y} \subset \mathbb{R}$	long-term outcome
$S(t) = S(T = t)$	potential short-term outcome of t
$Y(t) = Y(T = t)$	potential long-term outcome of t
$m \in \mathbb{Z}^+$	number of multi-tasks
$\mathcal{L}, \mathcal{R}$	objective functions of different tasks
θ, ξ, ζ	learnable parameters of different models
$\Psi(\cdot), \Phi(\cdot)$	representation networks for T and X
$\oplus$	operator for matrix concatenation
$h_s(\cdot), h_y(\cdot)$	hypothesis networks to estimate $S(t)$ and $Y(t)$
$\Pi(\cdot)$	deterministic policy network

by modeling it as an optimization problem. However, their exploration is currently limited to decision making for binary treatment.

3. Problem setup

3.1. Notations and assumptions

We aim to estimate both short-term and long-term treatment effects to determine an optimal treatment policy that balances these outcomes. As summarized in Table 1, let $\mathbb{D} = \{X_i, T_i, S_i, Y_i\}_{i=1}^n$ be the observational data for n units, where $X \in \mathcal{X} \subset \mathbb{R}^{d_X}$ denotes covariates, $T \in \mathcal{T} \subset \mathbb{R}$ is the continuous treatment, $S \in \mathcal{Y} \subset \mathbb{R}$ is the short-term outcome, and $Y \in \mathcal{Y} \subset \mathbb{R}$ is the long-term outcome. As illustrated in Fig. 1, X acts as a confounder affecting $\{T, S, Y\}$, while T directly influences S and Y .

Under the potential outcome framework (Rubin, 1974), we denote $S(t)$ and $Y(t)$ as the potential outcomes under treatment t . To identify these outcomes from observational data, we require the following standard assumptions (Cole & Frangakis, 2009; Kennedy et al., 2017):

1. *Consistency*. $S = S(T)$ and $Y = Y(T)$ for the observed treatment T .
2. *Unconfoundedness*. $S(t), Y(t) \perp T | X, \forall t \in \mathcal{T}$.
3. *Overlap*. $\mathbb{P}(T = t | X) > 0, \forall t \in \mathcal{T}$.
4. *Smoothness*. $S(t)$ and $Y(t)$ are differentiable with respect to t .

Based on these, the expected potential outcomes are identified as:

$$\mathbb{E}[S(t) | X = x] = \mathbb{E}[S | X = x, T = t], \quad (1)$$

$$\mathbb{E}[Y(t) | X = x, S = s] = \mathbb{E}[Y | X = x, T = t, S = s]. \quad (2)$$

Consistency ensures the observed outcome equals the potential outcome under the assigned treatment, implying the effect is independent of the administration mechanism (Cole & Frangakis, 2009). *Unconfoundedness* holds if X captures all variables simultaneously affecting $\{T, S, Y\}$, while *Overlap* ensures sufficient sample coverage across different treatment levels for stable estimation. For continuous treatments, *smoothness* is assumed to ensure the treatment effect exhibits a gradual, predictable trend, avoiding abrupt fluctuations from minor dosage changes (Kennedy et al., 2017). These assumptions are standard in real-world applications, such as public health (Doll & Hill, 1952), clinical regimes (Charig et al., 1986), and economics (Angrist & Krueger, 1991).

3.2. Preliminary

Pareto-Optimal Estimation (POE). Jointly estimating short- and long-term outcomes often encounters optimization conflicts, where gradients from a dominant task may bias the shared representation learning and degrade overall performance. To address this, we reformulate the estimation as a multi-task optimization problem. Let $\mathcal{L}(\theta) =$

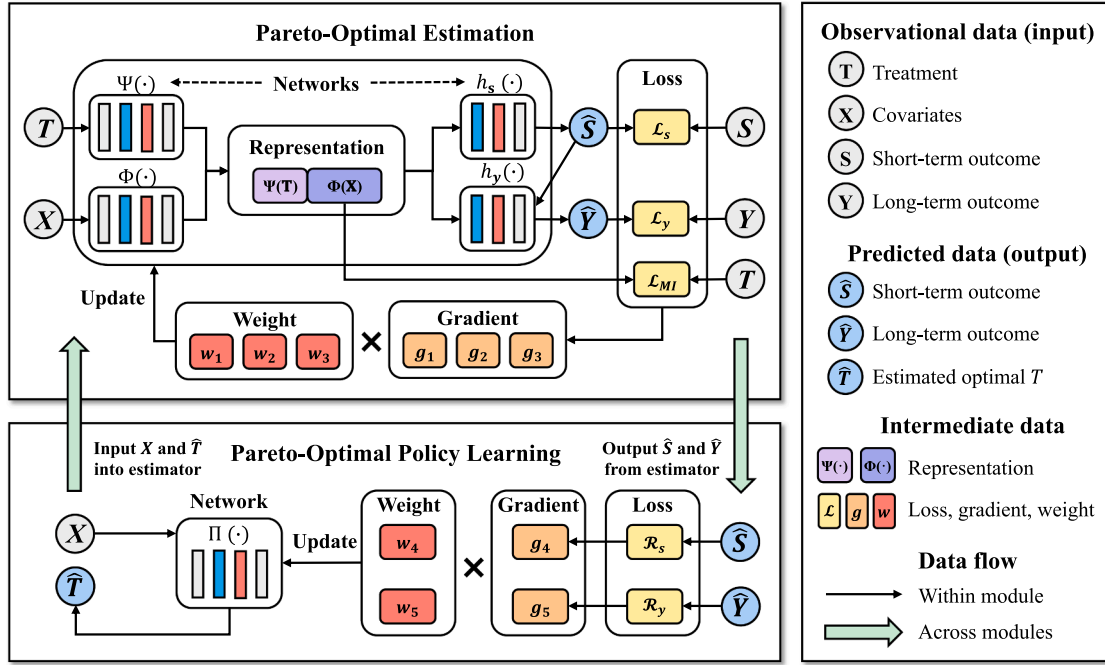


Fig. 2. The architecture of our model. Observational variables are marked in gray and the results predicted by our models are marked in blue. Intermediate data includes the learned representations ($\Psi(T)$ and $\Phi(X)$) and parameters used in model training (losses, gradients, and weights for three tasks). There are two modules, i.e., Pareto-Optimal Estimation and Pareto-Optimal Policy Learning. The data flow within a single module is represented by black thin arrows while the data exchange between two modules is depicted by thick green arrows.

$[\mathcal{L}_1(\theta), \dots, \mathcal{L}_m(\theta)]^\top$ denote the objective vector for m tasks ($m = 3$), including confounding balancing defined as Eq. (5) and outcome regressions defined as Eq. (7). Let θ denote the parameters of the overall backbone.

Due to conflicting optimization directions, a single global optimum θ^* that minimizes all objectives simultaneously is often unattainable. We thus treat it as a Pareto estimation problem (Lv et al., 2021; Momma et al., 2022; Zitzler & Thiele, 1999):

1. **Pareto dominance.** A solution θ' dominates another solution θ if $\forall i \in \{1, \dots, m\}, \mathcal{L}_i(\theta') \leq \mathcal{L}_i(\theta)$ and $\exists j \in \{1, \dots, m\}, \mathcal{L}_j(\theta') < \mathcal{L}_j(\theta)$. For simplicity, we denote θ' dominating θ as $\theta' \prec \theta$, and $\theta' \not\prec \theta$ otherwise.
2. **Pareto optimality.** The solution θ^* is Pareto optimal if there is no solution dominating it. Formally, $\theta \not\prec \theta^* \forall \theta \in \mathbb{R}^z - \{\theta^*\}$.
3. **Pareto frontier.** All the solutions that are Pareto-optimal collectively comprise of the Pareto frontier.

Following (Lv et al., 2021), we employ a continuous Pareto optimization algorithm to update θ , exploring the Pareto frontier to identify the optimal representation for counterfactual estimation.

Pareto-Optimal Policy Learning (POPL). Our ultimate goal is to identify an adaptive treatment strategy π that balances short-term rewards and long-term outcomes. Let $\mathbf{R}(\pi) = [\mathcal{R}_s(\pi), \mathcal{R}_y(\pi)]^\top$ be the reward vector, and the two rewards are implemented as regret loss Eqs. (12) and (13). Since S and Y often exhibit conflicting trends with respect to treatment T , we formulate policy learning as a multi-objective optimization problem to seek solutions along the Pareto frontier.

Formally, POPL aims to identify a set of Pareto-optimal policies such that for any policy π^* , no other feasible policy can improve one outcome without degrading the other. Unlike methods seeking a single fixed trade-off, POPL explores the entire Pareto frontier, delineating a spectrum of optimal solutions. This provides a flexible decision-making framework, allowing for the selection of treatment regimes calibrated to different clinical or practical priorities regarding immediate gains versus delayed benefits.

4. Methodology

Guided by the preliminary above, we propose a unified Pareto-Efficient architecture (Fig. 2) that integrates Pareto optimization into both counterfactual estimation and decision-making. The framework consists of: (1) Pareto-Optimal Estimation (POE), which mitigates gradient interference among confounder balancing and multi-outcome regressions; and (2) Pareto-Optimal Policy Learning (POPL), which explores the treatment space to identify policies on the Pareto frontier of short- and long-term rewards.

4.1. Pareto-optimal estimation

As shown in Fig. 2 (upper panel), POE aims to learn a shared representation $\Phi(X)$ that is both balanced (independent of T) and predictive for multiple outcomes. Unlike traditional methods that use fixed trade-off hyper-parameters, POE treats the following three objectives as a multi-task Pareto optimization problem.

4.1.1. Confounder balancing for continuous treatment

We use representation networks to learn $\Psi(T)$ and $\Phi(X)$, and then concatenate them into a vector $(\Psi(T) \oplus \Phi(X))$. In a counterfactual manner, we then regress the short- and long-term causal effects. To ensure unconfoundedness, we enforce $\Phi(X) \perp T$. Although Integral Probability Metrics (IPM) is widely-used in binary treatment setting via pairwise comparisons, it requires discretizations for continuous treatment range, which can lead to information loss. Hilbert-Schmidt Independence Criterion (HSIC) is also popular, but is sensitive to kernel choice and incurs higher computational cost due to kernel matrix operations. Compared with them, we employ mutual information (MI) (Cheng et al., 2020) as a principled, information-theoretic regularization metric (Cheng et al., 2020), which can be efficiently estimated via neural networks and integrated seamlessly with our Pareto optimization framework.

Specifically, we adopt a two-phase training strategy to learn the variational distribution and then estimate the mutual information. Firstly,

we fix the parameters of $\Phi(\cdot)$, and use them to approach the mean ($\mu = \mu_\theta(\Phi(X))$) and variance ($\text{Var} = \text{Var}_\theta(\Phi(X))$) of the variational distribution $q_\theta(\Phi(X))$, where $\mu_\theta(\cdot)$ and $\text{Var}_\theta(\cdot)$ are two learnable networks. Then, with the representation $\Phi(X)$ fixed, we minimize the log-likelihood to optimize networks $\mu_\theta(\cdot)$ and $\text{Var}_\theta(\cdot)$ to learn the variational approximation $q_\theta(T|\Phi(X))$:

$$\mathcal{L}_{LLD} = -\frac{1}{n} \sum_{i=1}^n \log q_\theta(t_i|\Phi(x_i)), \quad (3)$$

$$\log q_\theta(t_i|\Phi(x_i)) = \frac{\mu_\theta(\Phi(x_i)) - t_i}{\exp(\log \text{Var}_\theta(\Phi(x_i)))} - \log \text{Var}_\theta(\Phi(x_i)), \quad (4)$$

where $\mathcal{L}_{LLD}$ means the log likelihood function as an approximation function for probability distributions. To reduce the relevance between the representations and the treatment, we minimize the mutual information between them. In this phase, we fix the parameters of $\mu_\theta(\cdot)$ and $\text{Var}_\theta(\cdot)$, and then update the representation network $\Phi(X)$ to minimize the mutual information:

$$\mathcal{L}_{MI} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (\log q_\theta(t_i|\Phi(x_i)) - \log q_\theta(t_j|\Phi(x_i))), \quad (5)$$

where $\log q_\theta(t_i|\Phi(x_i))$ represents the conditional log-likelihood of the positive sample pair $(\Phi(x_i), t_i)$ and $q_\theta(t_j|\Phi(x_i))_{i \neq j}$ represents the negative sample pair $(\Phi(x_i), t_j)_{i \neq j}$. We minimize Eq. (5) to optimize independent balanced representations $\Phi(X)$, i.e., minimizing differences between the positive and negative sample pairs. Then we use them to achieve an unbiased treatment effect estimation. However, this comes with the trade-off of significant information loss or increased costs due to the stringent constraints involved. Thus, we expect to combine information from both short-term and long-term outcomes to supplement and enrich the information necessary to improve the shared representation framework.

4.1.2. Shared representation for both short-term and long-term outcomes

After obtaining the balanced representation $\Phi(X)$ and treatment embedding $\Psi(T)$, we use their concatenation $\Phi(X) \oplus \Psi(T)$ as a shared representation for outcome estimation. To prevent the single-dimensional T from being overshadowed by high-dimensional X , we ensure $\Psi(\cdot)$ is an invertible transformation that $\mathbb{P}(S, Y | \Phi(X), T) = \mathbb{P}(S, Y | \Psi(T) \oplus \Phi(X))$, preserving the treatment signal throughout the regression. We adopt a sequential prediction structure to capture the temporal dependency: the short-term outcome $\hat{s}$ is predicted first and then utilized as an auxiliary input for the long-term outcome $\hat{y}$. Specifically, we use a hypothesis network $h_s : \phi \times \psi \rightarrow s \subset \mathbb{R}$ to predict the short-term outcome, followed by another hypothesis network $h_y : \phi \times \psi \times s \rightarrow y \subset \mathbb{R}$ to estimate the long-term outcome using this shared concatenation representation $\Psi(T) \oplus \Phi(X)$. Formally,

$$\hat{s}_i = h_s(\Phi(x_i), \Psi(t_i)), \quad \hat{y}_i = h_y(\Phi(x_i), \Psi(t_i), \hat{s}_i). \quad (6)$$

We aim to minimize the mean squared error (MSE):

$$\mathcal{L}_s = \frac{1}{n} \sum_{i=1}^n (s_i - \hat{s}_i)^2, \quad \mathcal{L}_y = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (7)$$

where s_i and y_i are the true outcomes in the observational data. Ultimately, we can integrate these loss functions into the total objective function with three hyper-parameters $\mathbf{w} = \{\alpha, \beta, \gamma\}$:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{MI} + \beta \cdot \mathcal{L}_s + \gamma \cdot \mathcal{L}_y. \quad (8)$$

Existing methods typically rely on fixed hyper-parameters to manage the contribution of each single objective. However, such a manual weighting strategy often struggles when tasks exhibit conflicting optimization directions, leading to gradient interference that compromises the shared representation. To resolve these conflicts without exhaustive hyper-parameter tuning, we reformulate the joint estimation as a multi-objective optimization problem. This allows us to leverage Pareto optimization, introduced in the next section, to adaptively identify a common descent direction. It ensures an optimal balance between balancing constraints and outcome regressions (Lv et al., 2021; Momma et al., 2022; Zitzler & Thiele, 1999).

4.1.3. Pareto optimization

To resolve the aforementioned task conflicts, we reformulate the joint optimization into a continuous Pareto Optimization Problem (Lin et al., 2019; Lv et al., 2021; Ma et al., 2020). Let $\mathbf{G} = (g_1, g_2, g_3) = (\nabla_\xi \mathcal{L}_{MI}, \nabla_\xi \mathcal{L}_s, \nabla_\xi \mathcal{L}_y)$ the gradient matrix with respect to the shared parameters $\xi = \{\Psi, \Phi, h_s, h_y\}$. Due to the varying scales of different loss components, we apply a padding operator to ensure gradient consistency. Following MGDA (Sener & Koltun, 2018), we seek a weight vector $\mathbf{w} = [w_1, w_2, w_3]^\top = [\alpha, \beta, \gamma]^\top$ that finds a common descent direction to minimize all objectives simultaneously:

$$\min_{\mathbf{w}} J = \frac{1}{2} \left\| \sum_{i=1}^m w_i \nabla g_i(\xi) \right\|_2^2 = \frac{1}{2} \mathbf{w}^\top \nabla g(\xi) (\nabla g(\xi))^\top \mathbf{w} \quad (9)$$

$$s.t. \mathbf{w}^\top \mathbf{1}_m = 1, \quad \mathbf{w} \geq 0,$$

where m is the number of tasks ($m = 3$ in POE), and $\mathbf{w}$ represents the weights to adaptively trade-off among them. Moreover, each $g_i(\cdot)$ refers to one learning objective, i.e., a gradient function for each loss. The term $\mathbf{G}^\top \mathbf{G}$ represents the Gram matrix of gradients, capturing the inter-task correlations. To efficiently solve this constrained optimization, we employ an augmented Lagrangian method to transform Eq. (9) into an unconstrained objective:

$$J = \frac{1}{2} \mathbf{w}^\top \nabla g(\xi) (\nabla g(\xi))^\top \mathbf{w} + \mu (\mathbf{w}^\top \mathbf{1}_m - 1) + \frac{\rho}{2} \|\mathbf{w}^\top \mathbf{1}_m - 1\|_2^2, \quad (10)$$

where μ and ρ are the Lagrangian and augmented Lagrangian coefficients, respectively. The optimal weights $\mathbf{w}^*$ can then be analytically updated by:

$$\begin{cases} \mathbf{w} = \max \left(0, (\nabla g(\xi) (\nabla g(\xi))^\top + \rho \mathbf{I})^{-1} (\rho \mathbf{I} - \mu \mathbf{I}) \right), \\ \mu \leftarrow \mu + \rho (\mathbf{w}^\top \mathbf{1}_m - 1). \end{cases} \quad (11)$$

The cumulative update direction in each iteration is given by $\mathbf{d} = \mathbf{G}\mathbf{w}^*$. The shared parameters ξ are then updated via gradient descent: $\xi^{i+1} = \xi^i - \eta \mathbf{d}$, where η is the step size. This adaptive weighting mechanism, summarized in Algorithm 1, ensures that the shared representation $\Phi(X)$ is optimized along the Pareto frontier, effectively balancing the trade-off between confounder removal and multi-outcome predictive accuracy.

Algorithm 1 Pareto-optimal estimation.

Input: Initial parameters ξ , step size η , maximum iteration number K

- 1: **for** $i \leftarrow 1$ **to** K **do**
- 2: Calculate the losses $\mathcal{L}_{MI}$, $\mathcal{L}_s$, and $\mathcal{L}_y$ by Eqs. (5) and (7)
- 3: Calculate $[g_1, g_2, g_3] \leftarrow [\nabla_\xi \mathcal{L}_{MI}, \nabla_\xi \mathcal{L}_s, \nabla_\xi \mathcal{L}_y]$
- 4: Obtain $\mathbf{w}^*$ by Eq. (11)
- 5: Calculate optimization direction by $\mathbf{d} \leftarrow \mathbf{G}\mathbf{w}^*$
- 6: Update parameters by $\xi^{i+1} \leftarrow \xi^i - \eta \mathbf{d}$
- 7: **end for**

Output: Updated parameters ξ^K

4.2. Pareto-optimal policy learning

Building upon the counterfactual estimates provided by the POE module in Section 4.1, we aim to identify an adaptive treatment policy $\Pi(X) : \mathcal{X} \rightarrow \mathcal{T}$ that maximizes both short- and long-term rewards. In practice, decision-makers often face dilemmas where a treatment dosage T may benefit one outcome while compromising the other. To address this, we formulate policy learning as a multi-objective optimization problem to explore the Pareto frontier of treatment effects. Given the pre-treatment covariates X , the deterministic policy network $\Pi(\cdot)$ is trained to minimize the regret relative to the maximum achievable rewards in the observed domain. Formally, we define the regret losses for

short-term outcomes as:

$$\mathcal{R}_s = \frac{1}{N} \sum_{i=1}^N \min_{t_i} \{S^* - h_s(\Phi(x_i), \Psi(\Pi(x_i)))\}, \quad (12)$$

where S^* denotes the reference maximum outcomes (or target rewards) in $\mathbb{D}$. Similarly, the objective with respect to the long-term outcome is:

$$\mathcal{R}_y = \frac{1}{N} \sum_{i=1}^N \min_{t_i} \{Y^* - h_y(\Phi(x_i), \Psi(\Pi(x_i)), \hat{h}_s)\}. \quad (13)$$

Algorithm 2 Pareto-optimal policy learning.

Input: Initial parameters ζ , step size λ , maximum iteration number N

- 1: **for** $i \leftarrow 1$ **to** N **do**
- 2: Calculate the losses $\mathcal{R}_s$ and $\mathcal{R}_y$ by Eqs. (12) and (13)
- 3: Calculate $[g_4, g_5] \leftarrow [\nabla_{\zeta} \mathcal{R}_s, \nabla_{\zeta} \mathcal{R}_y]$
- 4: Obtain $\mathbf{w}^*$ by Eq. (11)
- 5: Calculate optimization direction by $\mathbf{d} \leftarrow \mathbf{G}\mathbf{w}^*$
- 6: Update parameters by $\zeta^{i+1} \leftarrow \zeta^i - \lambda \mathbf{d}$
- 7: **end for**

Output: Updated parameters ζ^N

Similar to the estimation phase, these two regret objectives may be non-aligned or conflicting. We therefore employ the continuous Pareto optimization algorithm (detailed in Algorithm 2) to update the policy parameters. By adaptively reconciling the gradients $\nabla \mathcal{R}_s$ and $\nabla \mathcal{R}_y$, POPL ensures that the learned policy $\Pi(\cdot)$ converges toward the Pareto frontier, providing a spectrum of optimal treatment regimes that satisfy different trade-off preferences between immediate and delayed benefits.

Algorithm 3 Overall workflow.

Input: Dataset $\mathbb{D}$, initial parameters of POE ξ , initial parameters of POPL ζ , step size η , maximum iteration K

- 1: **for** $i \leftarrow 1$ **to** K **do**
- 2: inputs $\leftarrow [\Phi(\mathbb{D}.X), \Psi(\mathbb{D}.T)]$
- 3: $\hat{S} \leftarrow h_s(\text{inputs})$ by Eq. (6)
- 4: $\hat{Y} \leftarrow h_y(\text{inputs}, \hat{S})$ by Eq. (6)
- 5: $\mathcal{L}_{MI} \leftarrow MI(\Phi(X), T)$ by Eq. (5)
- 6: $\mathcal{L}_s \leftarrow MSE(\hat{S}, \mathbb{D}.S)$ by Eq. (7)
- 7: $\mathcal{L}_y \leftarrow MSE(\hat{Y}, \mathbb{D}.Y)$ by Eq. (7)
- 8: $\xi \leftarrow \xi'$ after $\mathcal{L}.\text{backward}()$
- 9: **end for**
- 10: $\xi \leftarrow \text{POE.train}(\xi, \eta, K)$ by Algorithm 1
- 11: $[s^*, y^*] \leftarrow [\max(\mathbb{D}.S), \max(\mathbb{D}.Y)]$
- 12: **for** $i \leftarrow 1$ **to** K **do**
- 13: $\hat{T} \leftarrow \Pi(\mathbb{D}.X)$
- 14: $\hat{S} \leftarrow h_s(\Phi(\mathbb{D}.X), \hat{T})$ by Eq. (6)
- 15: $\hat{Y} \leftarrow h_y(\Phi(\mathbb{D}.X), \hat{T}, \hat{S})$ by Eq. (6)
- 16: $\mathcal{R}_s \leftarrow \text{Regret}(\hat{S}, s^*)$ by Eq. (12)
- 17: $\mathcal{R}_y \leftarrow \text{Regret}(\hat{Y}, y^*)$ by Eq. (13)
- 18: $\zeta \leftarrow \eta'$ after $\mathcal{R}.\text{backward}()$
- 19: **end for**
- 20: $\zeta \leftarrow \text{POPL.train}(\zeta, \eta, K)$ by Algorithm 2

Output: Updated parameters ξ, ζ

4.3. Overall

The complete workflow is summarized in Algorithm 3. The framework proceeds in two integrated stages including estimation and policy learning. In POE, given observational data, the representation networks $\{\Psi, \Phi\}$ project treatment T and covariates X into a balanced latent space. These representations are fed into hypothesis networks h_s and h_y to generate counterfactual estimates $\{\hat{S}, \hat{Y}\}$. Specifically, $\hat{S}$ is incorporated as an input for $\hat{Y}$ to capture temporal dependencies. The Pareto

Table 2

Statistics of datasets.

Dataset	Treatment	# train	# test	$\dim_x$
Crime	binary	171	43	27
MIMIC-III	binary	4,293	1,840	69
IHDP	continuous	537	135	25
Jobs	continuous	2,056	514	17
Twins	continuous	3,795	949	38
Simulation	continuous	16,000	4,000	2

module reconciles three potentially conflicting objectives: the balancing loss $\mathcal{L}_{MI}$, and the regression losses $\{\mathcal{L}_s, \mathcal{L}_y\}$. By calculating gradients $\mathbf{G}$ and adaptively updating weights $\mathbf{w}$, the module identifies a common descent direction that optimizes all objectives toward the Pareto frontier of estimation accuracy and confounder removal.

Building upon the fixed estimator, POPL trains a policy network $\Pi(X)$ to map covariates to optimal treatment values $\hat{T}$. The predicted $\hat{T}$ is evaluated through the pre-trained POE module to obtain expected outcomes $\{\hat{S}, \hat{Y}\}$. We define regret losses $\{\mathcal{R}_s, \mathcal{R}_y\}$ as the discrepancy between these predictions and the maximum potential rewards. A secondary Pareto optimization process is applied to reconcile the gradients of these regrets, ensuring the learned policy $\Pi(\cdot)$ explores the optimal trade-off space and converges to a Pareto-optimal treatment regime.

5. Experiments

5.1. Datasets

There are totally 6 datasets used in the evaluation, including 2 real-life dataset (Crime and MIMIC-III), 3 semi-synthetic datasets (IHDP, Jobs, and Twins), and 1 simulation dataset (Simulation). We summarize their statistics in Table 2. Generally speaking, it is more challenging to make counterfactual predictions if there is a smaller number of samples in the training set.

Crime.² Some researchers have studied the impact of New York's bail reform (Zhou et al., 2023), which is implemented in January 1, 2020. The raw data record the aggregate level of a specific crime in 27 cities for each day from January 1, 2018 to March 15, 2020. We convert crime data for 805 days into monthly average crime rates (normalized after dividing by the population). We use data from all months prior to January 1, 2020 as covariates. Including city names and crime categories (two-level classification), we obtain a total of 27 covariates. The average crime rates in January and March in 2020 are treated as short- and long-term outcomes, respectively. Note that the treatment in this dataset is binary. Only the crimes in New York belong to the treated group, and the rest consist of the control group.

MIMIC-III.³ It is a comprehensive, publicly available database containing de-identified health data from patients admitted to critical care units in a large tertiary care hospital (Johnson et al., 2016). Referring to data processing in the Causal Transformer (Melnichuk et al., 2022), we obtain 69 covariates on the vital signs of patients. We choose whether to take mechanical ventilation during the first 5 time steps as T , blood pressure in the 10th time step as S , and blood pressure in the 15th time step as Y .

IHDP.⁴ It is a commonly adopted benchmark dataset collected from the Infant Health and Development Program (Brooks-Gunn et al., 1992). It is a longitudinal study conducted in the United States from 1985 to 1993, with the aim of determining how low birth weight and premature birth affect the future development of infants (measured by cognitive

² Data are available at <https://tandf.figshare.com/ndownloader/articles/24262352/versions/1>.

³ Data and codes for processing are available at https://github.com/MLforHealth/MIMIC_Extract.

⁴ Data are available at <https://www.fredjo.com/>.

test score). There are 25 covariates covering various aspects of the infants together with their mothers, such as the neonatal health index, prenatal care, mother's age, education, etc. We utilize these covariates, denoted as X , to generate the continuous treatments by

$$T = \sum_{i=1}^{25} \cos(1 + X_i^2). \quad (14)$$

The short-term and long-term outcomes are designed as

$$S = 2.5 \sin(2 + T) + 0.25 \sum_{i=1}^{25} e^{-X_i^2} + 1.25, \quad Y = 0.1T^2 - \log(S) + 2 \sum_{i=1}^{25} X_i + 5. \quad (15)$$

Jobs. This dataset (LaLonde, 1986) comprises of the data from two sources, i.e., Lalonde experiment and the Panel Study of Income Dynamics (PSID). The download link is consistent with the IHDP dataset, which is available in the footnote. This study focuses on how job training could affect an individual's employment status. Information such as age, education, ethnicity, and previous earnings is included in X . We use the 17 covariates to generate the treatment variables by the following equation:

$$T = 0.2 \sum_{i=1}^{17} (\sin(X_i) + e^{-X_i^2}). \quad (16)$$

Similar to the generation formulations utilized in IHDP, the two outcome variables in Jobs dataset are defined as:

$$S = 1.7 \sin(2T) + 0.05 \sum_{i=1}^{17} X_i + 3.4, \quad Y = 0.7T - S + 0.02 \log(1 + X_i^2) + 5. \quad (17)$$

Twins.⁵ It collects all births in the USA between 1989–1991, and only twins weighing less than 2kg are recorded without missing features (Almond et al., 2005). Covariates measure information in 38 dimensions, including pregnancy, quality of care, pregnancy risk factors, residence, etc. Birth weight is regarded as the treatment ($T = 1$ for the heavier one in the twin and $T = 0$ for the other). The outcome of interest is 1-year mortality. In our experiment, we generate the treatment with the help of 38 covariates through

$$T = 0.5 \sum_{i=1}^{38} \log(1 + e^{X_i}) - 15. \quad (18)$$

The expressions for short-term and long-term outcomes are given below.

$$S = 0.75 \sin T + 0.02 \sum_{i=1}^{38} e^{-X_i^2} + 2, \quad Y = 0.2e^{\sqrt{T}} - 0.2 \cos S + 0.001 \sum_{i=1}^{38} X_i^2 + 2. \quad (19)$$

Simulation. Considering the diversity in the number of covariates and samples, we also generate a simulated dataset of 20,000 samples with 2 covariates. Each dimension of X follows a uniform distribution, i.e., $x_i \sim U(0, 2)$. Treatments are assigned by

$$T = \sum_{i=1}^2 \log(1 + e^{X_i}). \quad (20)$$

The short-term and long-term outcomes are defined as:

$$S = 0.4 \sin T + 0.2 \sum_{i=1}^2 e^{-X_i^2} + 1, \quad Y = 0.1e^{\sqrt{T}} - 0.1 \cos S + 0.01 \sum_{i=1}^2 X_i^2 + 1. \quad (21)$$

Note that X are quite different among the three semi-synthetic datasets and the simulated dataset, and we apply different equations to generate treatments and outcomes so as to guarantee the conflicts between S and Y .

5.2. Baselines

Considering that multi-task learning techniques are underexplored in the estimation of long-term causal effects, we choose the relevant methods as baselines from the following areas in Table 3.

Causal inference of continuous treatment. We use these backbones to train two networks, separately predicting S and Y .

- **TARNet** (Shalit et al., 2017). It is a classic model applied in the binary treatment setting. There is a representation network to learn a shared embedding of X between the treated group and the control group, together with two hypothesis networks to predict the outcomes of each group. We extend it into address continuous treatment by substituting two hypothesis networks with a common one shared with different groups.
- **CFR** (Shalit et al., 2017). We develop TarNet into CFR by adding a mutual information module between T and the representation of X . The loss calculating such mutual information is exactly the representation loss designed in our model, whose formal definition is given in Eq. (5).
- **DRNet** (Schwab et al., 2020). There is also a base layer to learn the representation of X , and several treatment layers to learn the specific representations of each kind of T . For each treatment layer, multiple heads are nested to divide the treatment domain $\mathcal{T}$ into multiple intervals. The corresponding dosage function for each interval is learned.
- **VCNet** (Nie et al., 2021). The representation of X is extracted as Z through a non-linear mapping, where a propensity estimation in Dragonnet (Shi et al., 2019) is applied to ensure the distribution balance among various treatment values. Furthermore, there is only one prediction head with varying coefficients for counterfactual prediction given T and Z .
- **CRNet** (Liu et al., 2020). It demonstrates the importance of balancing and prognostic representations, that is, $T, Y \perp X \mid \Phi(X)$. It learns a double balancing representation with such constraints via contrastive learning while preserving the continuity of treatments.

Long-term policy learning. There are two relevant works that focus on balancing between S and Y . However, they can only address the problem of decision-making with binary treatment. Short-long-term-rewards (Wu et al., 2024) models the trade-off between S and Y as an optimization problem. Based on this, it proposes a principled policy learning approach. Decomposition-based policy learning (DPPL) (Yang et al., 2024a) transforms the entire optimization problem into multiple subproblems to overcome the limitations of the traditional linear weighting method. We only apply them in the evaluation of estimation accuracy on two real-life datasets where treatment is binary. Note that the estimation modules about S and Y are the same in these two works.

Multi-task learning. FairGrad (FG) (Ban & Ji, 2024) is one of the state-of-the-art methods in multi-task learning. It formulates the optimization issue as a α -fair utility maximization problem. It achieves fair optimization among different tasks by maximizing the total utility of the task loss decrease rates. We apply FairGrad as our baseline by regarding the estimation of S and Y as two tasks for joint optimization.

5.3. Results

5.3.1. Estimation

We uniformly select 50 points within the possible interval of $\mathcal{T}$ as t^{cf} for counterfactual predictions, and use Mean Squared Error (MSE) to evaluate the ability of all the models that are mentioned above. After training the models with 10 randomly picked seeds, all the results are reported in the form of (mean value $\pm$ standard deviation).

Experiment results of three semi-synthetic datasets are concluded in Table 4, providing a clear comparison between our method and six baselines in terms of the estimation performance. Our method consistently outperforms all baselines in various datasets. In the IHDP dataset,

⁵ Data are available at <https://www.nber.org/research/data/linked-birthinfant-death-cohort-data>.

Table 3
Details of baselines.

Area	Method	T	Open-source
Causal Inference	TARNet	$\mathbb{R}$	https://github.com/clinicalml/cfnet
	CFR	$\mathbb{R}$	
	DRNet	$\mathbb{R}$	https://github.com/d909b/drnet
	VCNet	$\mathbb{R}$	https://github.com/lushleaf/varying-coefficient-net-with-functional-tr
Policy Learning	CRNet	$\mathbb{R}$	https://github.com/euzmin/Contrastive-Balancing-Representation-Network-CRNet
	SLTR	$\{0, 1\}$	https://github.com/YanaZeng/Short_long_term-Rewards
	DPPL	$\{0, 1\}$	supplemental link from NeurIPS
MTL	FairGrad	$\mathbb{R}$	https://github.com/OptMN-Lab/fairgrad

Table 4
Estimation results of semi-synthetic datasets (IHDP, Jobs, Twins).

	IHDP		Jobs		Twins	
	MSE _s	MSE _y	MSE _s	MSE _y	MSE _s	MSE _y
TarNet	1.434 $\pm$ 0.031	0.297 $\pm$ 0.051	1.308 $\pm$ 0.048	8.561 $\pm$ 1.279	0.583 $\pm$ 0.024	0.348 $\pm$ 0.014
CFR	0.461 $\pm$ 0.050	0.243 $\pm$ 0.102	1.001 $\pm$ 0.318	1.038 $\pm$ 0.349	0.071 $\pm$ 0.023	0.036 $\pm$ 0.018
DRNet	3.284 $\pm$ 0.668	0.585 $\pm$ 0.182	1.874 $\pm$ 0.114	7.633 $\pm$ 1.249	0.411 $\pm$ 0.203	0.286 $\pm$ 0.137
VCNet	1.636 $\pm$ 0.060	0.397 $\pm$ 0.111	1.754 $\pm$ 0.023	6.881 $\pm$ 0.606	0.177 $\pm$ 0.112	0.155 $\pm$ 0.063
CRNet	0.733 $\pm$ 0.084	0.657 $\pm$ 0.196	0.550 $\pm$ 0.401	0.739 $\pm$ 0.324	0.086 $\pm$ 0.088	0.072 $\pm$ 0.028
FG	2.791 $\pm$ 0.359	0.216 $\pm$ 0.048	0.500 $\pm$ 0.187	0.487 $\pm$ 0.309	0.150 $\pm$ 0.084	0.017 $\pm$ 0.004
Ours	0.382* $\pm$ 0.025	0.182* $\pm$ 0.024	0.238* $\pm$ 0.096	0.185* $\pm$ 0.063	0.041* $\pm$ 0.001	0.015* $\pm$ 0.001

* means statistically significant results (p-value ≤ 0.05).**Table 5**
Estimation results of Simulation and real-word datasets (Crime, MIMIC-III).

	Simulation		Crime		MIMIC-III	
	MSE _s	MSE _y	MSE _s	MSE _y	MSE _s	MSE _y
TarNet	0.442 $\pm$ 0.028	0.206 $\pm$ 0.019	4.322 $\pm$ 2.494	5.221 $\pm$ 1.792	1.004 $\pm$ 0.011	0.923 $\pm$ 0.013
CFR	0.222 $\pm$ 0.211	0.088 $\pm$ 0.134	0.911 $\pm$ 0.296	1.346 $\pm$ 0.290	0.862 $\pm$ 0.041	0.808 $\pm$ 0.028
DRNet	0.515 $\pm$ 0.093	0.216 $\pm$ 0.029	5.387 $\pm$ 1.104	3.256 $\pm$ 1.459	1.003 $\pm$ 0.012	0.924 $\pm$ 0.010
VCNet	0.293 $\pm$ 0.017	0.184 $\pm$ 0.008	1.155 $\pm$ 0.498	2.639 $\pm$ 1.511	1.025 $\pm$ 0.013	0.911 $\pm$ 0.017
CRNet	0.031 $\pm$ 0.012	0.041 $\pm$ 0.020	1.396 $\pm$ 0.491	1.941 $\pm$ 0.826	1.001 $\pm$ 0.005	0.918 $\pm$ 0.002
DPPL	–	–	1.914 $\pm$ 1.161	2.357 $\pm$ 1.216	1.015 $\pm$ 0.090	0.948 $\pm$ 0.094
FG	0.052 $\pm$ 0.019	0.018 $\pm$ 0.013	2.984 $\pm$ 1.056	2.669 $\pm$ 0.837	0.791 $\pm$ 0.003	0.750 $\pm$ 0.003
Ours	0.012* $\pm$ 0.014	0.005* $\pm$ 0.006	0.838* $\pm$ 0.149	0.510* $\pm$ 0.078	0.775* $\pm$ 0.003	0.734* $\pm$ 0.004

* means statistically significant results (p-value ≤ 0.05).

although the CFR method shows competitive performance with relatively low MSE_s (0.461 $\pm$ 0.050) and MSE_y (0.243 $\pm$ 0.102), our method still achieves the lowest MSE_s (0.382 $\pm$ 0.025) and MSE_y (0.182 $\pm$ 0.024). In the case of Jobs dataset, our method surpasses the second-best approach by a large margin of 0.262 in MSE_s and 0.302 in MSE_y, indicating our model's superiority in terms of accuracy and stability. The Twins dataset is less challenging because of the sufficient number of samples for training. This situation is quite similar to that of the Simulation dataset in Table 5. However, our approach still outperforms all baselines.

There are two real-word datasets in Table 5, where the treatment is binary. Therefore, we additionally apply two long-term policy learning baselines (Wu et al., 2024; Yang et al., 2024a) here. Since they share the same estimation module, we combine them into a single line called DPPL for reporting. As for the Crime dataset, only the crimes in New York City are regarded as the treated group (implementing bail reform). For each sample in the treated group, we search for the three most similar samples with the highest Pearson correlation coefficients from the control group.

The formal definition of correlation coefficient between unit i and j is

$$\rho(X_i, X_j) = \frac{\text{cov}(X_i, X_j)}{\sigma(X_i)\sigma(X_j)}, \quad (22)$$

where $\rho(X_i, X_j)$ means their covariance, and $\sigma(X_i)$ and $\sigma(X_j)$ are the standard deviations. Afterward, we calculate a weighted average of the selected samples' outcomes as the ground truth for counterfactual prediction. Based on the reported results, our POE has an obvious margin over the other baselines. In another real-world MIMIC-III dataset, our method still stands out, while FairGrad and CFR also perform well.

In summary, our model outperforms all the baselines in terms of predicting both the short-term and long-term outcomes on all datasets. It also demonstrates significant improvements in the majority of cases. For the few scenarios where the results are not statistically significant, it is primarily due to the high variance exhibited by certain baselines. The superiority of our model in MSE_y is easy to explain. Unlike the baselines that train two separate networks to predict S and Y independently, we leverage the predicted value $\hat{S}$, which is derived from the first prediction head, to inform the prediction of Y in the second head. By incorporating this additional information, our model achieves more accurate estimations of the causal effect on Y . Furthermore, our model also demonstrates superior performance in predicting short-term outcomes. This can be attributed to the fact that the long-term outcome Y contains information from both the covariates X and the short-term outcome S . Therefore, by optimizing the prediction of Y , our model effectively enhances the representation learning of X and improves the prediction accuracy of S in turn.

5.3.2. Policy learning

Visualization of policy learning in different datasets is demonstrated in Fig. 3. The x-axis corresponds to short-term outcomes S , the y-axis represents values of the assigned treatment T , and the z-axis refers to long-term outcomes Y . Setting the coordinate axes in this way gives us a more clear insight of the conflicting relationship between S (x-axis) and Y (z-axis). Here we applied four datasets with ground truth of continuous treatment effects, which are convenient for generating counterfactual outcomes and drawing the Pareto frontier during evaluation.

For each dataset, we randomly select two samples as representatives and plot the three-dimensional visualization of the decision-making. Specifically, we discretize the interval $\mathcal{T}$ of treatment by uniform sampling and regard the sampled points as possible values t^{cf} for counterfactual prediction. Afterwards, we iterate on all these t^{cf} , and use the generation formulas (mentioned in Section 5.1) to obtain the corresponding s^{cf} and y^{cf} . These counterfactual outcomes are depicted as gray points in Fig. 3. Larger values for S and Y are preferred in our setting, and thus the Pareto frontier of these points is the upper right boundary. We have highlighted it in blue. The optimal treatment t^* for a sample is predicted by the policy learning module $\Pi(x_i)$ in Section 4.2. We mark the data point (s^*, t^*, y^*) as a red star in Fig. 3. The red star is always located on the Pareto frontier. This means that the t^* determined by our model will not be dominated by the outcomes

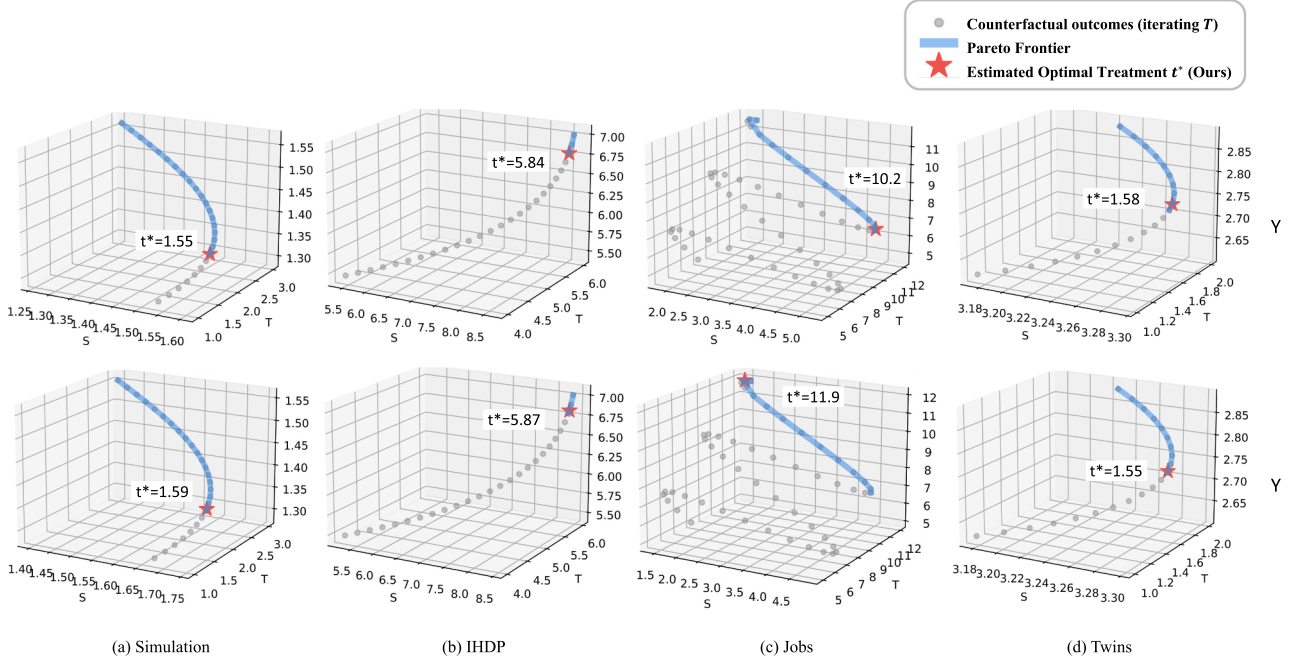


Fig. 3. Visualization of policy learning, and higher values of S and Y are preferred. The optimal value t^* estimated by our model is always located at the Pareto frontier. The x-axis represents the value of short-term outcome S , y-axis referring to T and z-axis is long-term outcome Y . This figure depicts multi-outcomes for $t \in [1, 3]$, $[4, 6]$, $[5, 12]$, and $[1, 2]$ on Simulation, IHDP, Jobs, and Twins dataset, respectively.

of any other values for T . Fig. 3 is of high interpretability for various cases with different treatment dosages and how to make decisions accordingly.

5.3.3. Ablation study

We conducted the ablation study in both simulated and semi-synthetic datasets, whose results are demonstrated in Tables 6 and 7. There are 5 sub-modules in POE:

- Module 1 abbreviated as $\mathcal{L}_s$. It is a prediction head for S , whose objective is to minimize the estimation error $\mathcal{L}_s$ defined in Eq. (7).
- Module 2 abbreviated as $\mathcal{L}_y$. It refers to a prediction head for Y , aiming to minimize the estimation loss $\mathcal{L}_y$ introduced in Eq. (7).
- Module 3 abbreviated as $\mathcal{L}_{MI}$. It means learning a balanced representation for continuous treatment via mutual information constraints. Its objective is to minimize $\mathcal{L}_{MI}$ in Eq. (5).
- Module 4 abbreviated as $\hat{S}$. It can be regarded as an extended version of module 2, but using the estimated value of S derived from module 1 as an additional input.
- Module 5 abbreviated as Pareto. It is the Pareto optimization process for model updates, which is introduced in Section 4.1.3.

The first two line in Tables 6 and 7 correspond to the CFR models that separately predict S or Y . The results reported in these two rows are the same as those in Tables 4 and 5. For comparison, the fourth line refers to an end-to-end framework that jointly trains multiple tasks ($\mathcal{L}_s, \mathcal{L}_y, \mathcal{L}_{MI}$) with predefined preferences. In our statement in Section 5.3.1, outcome Y contains information about S and X . Therefore, it can facilitate mutual enhancement to some extent if multiple tasks are learned jointly. The improvements in both MSE_s and MSE_y validate our viewpoint, especially the increases in Jobs, Twins, and Simulation datasets. Taking Simulation for example, there is a 76.6% increase in accuracy for S and 73.9% for Y .

To validate the effectiveness of the representation balance for continuous treatment, i.e. module 3 mentioned above, we could compare the results of the fourth row against the third row in the two tables. We can also refer to the results reported in the last line compared with those in the previous line. It can be seen that this strategy brings about a cer-

tain degree of improvement in each dataset that corresponds to different contexts.

Additionally, we found that utilizing the predicted value of short-term outcome, i.e., $\hat{S}$, as an input could facilitate the prediction of Y . It validates the contribution of module 4. Considering that Y already contains a part of the information from S in joint optimization, the improvement for Y achieved by using S as an additional input is not very significant. But the highest improvement of predicting Y still achieves 17.59% in Jobs. However, this approach means imposing stricter constraints on the accuracy of predicting S , leading to 55.8% improvement in the simulated dataset.

Actually, the method in the fifth line with modules 1–4 provides an initial solution to explore the Pareto frontier of our ultimate model. The complete version of our POE is the last line in Tables 6 and 7, i.e. applying Pareto optimization to trade off between all the aforementioned objectives. Although model performance had already converged to a local optimum, Pareto optimization successfully broke through this limitation and brought about the most significant performance improvement among all modules. To be specific, the decreases in the estimation errors of (S, Y) reach (45.5%, 55.6%) in the Jobs dataset. As for the simulated dataset, the complete version of our proposed method outperforms the sub-optimal one with an improvement of 47.8% and 77.3% in estimating S and Y , respectively.

According to the analysis above, the MSE error exhibits a monotonically decreasing pattern when analyzed row-wise. It validates the effectiveness of each newly introduced module, including joint optimization between multiple objectives, mutual information constraints for balanced representation of continuous treatment, information from $\hat{S}$ as input to predict Y , and Pareto optimization for further trade-off among conflicting objectives. Overall, a significant increase in performance is the end-to-end framework designed to jointly optimize multiple tasks. Based on this, we achieve the most substantial improvement through Pareto optimization among the three objectives.

5.3.4. Hyper-parameter analysis

According to Eq. (8), the objective function of POE is defined as $\mathcal{L} = \alpha \cdot \mathcal{L}_{MI} + \beta \cdot \mathcal{L}_s + \gamma \cdot \mathcal{L}_y$, where hyper-parameters $\{\alpha, \beta, \gamma\}$ are used to control confounding balance, short-term outcome pre-

Table 6
Results of ablation study on IHDP and Jobs datasets.

$\mathcal{L}_s$	$\mathcal{L}_y$	$\mathcal{L}_{MI}$	$\hat{S}$	Pareto	IHDP		Jobs	
					MSE _s	MSE _y	MSE _s	MSE _y
✓		✓			0.461 $\pm$ 0.050	–	1.001 $\pm$ 0.318	–
	✓	✓			–	0.243 $\pm$ 0.102	–	1.038 $\pm$ 0.349
✓	✓				0.441 $\pm$ 0.028	0.231 $\pm$ 0.016	0.533 $\pm$ 0.057	0.512 $\pm$ 0.063
✓	✓	✓			0.425 $\pm$ 0.020	0.220 $\pm$ 0.136	0.519 $\pm$ 0.192	0.506 $\pm$ 0.192
✓	✓	✓	✓		0.414 $\pm$ 0.034	0.195 $\pm$ 0.016	0.437 $\pm$ 0.155	0.417 $\pm$ 0.186
✓	✓	✓		✓	0.398 $\pm$ 0.016	0.197 $\pm$ 0.018	0.250 $\pm$ 0.127	0.217 $\pm$ 0.035
✓	✓	✓	✓	✓	0.382$\pm$0.025	0.182$\pm$0.024	0.238$\pm$0.096	0.185$\pm$0.063

Table 7
Results of ablation study on twins and simulation datasets.

$\mathcal{L}_s$	$\mathcal{L}_y$	$\mathcal{L}_{MI}$	$\hat{S}$	Pareto	Twins		Simulation	
					MSE _s	MSE _y	MSE _s	MSE _y
✓		✓			0.071 $\pm$ 0.023	–	0.222 $\pm$ 0.211	–
	✓	✓			–	0.036 $\pm$ 0.018	–	0.088 $\pm$ 0.134
✓	✓				0.064 $\pm$ 0.008	0.025 $\pm$ 0.001	0.057 $\pm$ 0.027	0.028 $\pm$ 0.007
✓	✓	✓			0.057 $\pm$ 0.025	0.024 $\pm$ 0.019	0.052 $\pm$ 0.053	0.023 $\pm$ 0.021
✓	✓	✓	✓		0.048 $\pm$ 0.003	0.018 $\pm$ 0.002	0.023 $\pm$ 0.023	0.022 $\pm$ 0.028
✓	✓	✓		✓	0.048 $\pm$ 0.006	0.019 $\pm$ 0.004	0.018 $\pm$ 0.012	0.009 $\pm$ 0.003
✓	✓	✓	✓	✓	0.041$\pm$0.001	0.015$\pm$0.001	0.012$\pm$0.014	0.005$\pm$0.006

Table 8
Results of hyper-parameter study on IHDP and jobs datasets.

(β, γ)	IHDP		Jobs	
	MSE _s	MSE _y	MSE _s	MSE _y
(0.1, 0.9)	1.222 $\pm$ 0.182	0.085 $\pm$ 0.015	0.291 $\pm$ 0.165	0.231 $\pm$ 0.127
(0.3, 0.7)	0.661 $\pm$ 0.059	0.136 $\pm$ 0.034	0.284 $\pm$ 0.100	0.240 $\pm$ 0.063
(0.5, 0.5)	0.441 $\pm$ 0.043	0.188 $\pm$ 0.028	0.393 $\pm$ 0.161	0.341 $\pm$ 0.143
(0.7, 0.3)	0.402 $\pm$ 0.031	0.186 $\pm$ 0.032	0.485 $\pm$ 0.145	0.423 $\pm$ 0.152
(0.9, 0.1)	0.386 $\pm$ 0.032	0.196 $\pm$ 0.028	0.520 $\pm$ 0.161	0.485 $\pm$ 0.181
Ours	0.382$\pm$0.025	0.182$\pm$0.024	0.238$\pm$0.096	0.185$\pm$0.063

Table 9
Results of hyper-parameter study on twins and simulation datasets.

(β, γ)	Twins		Simulation	
	MSE _s	MSE _y	MSE _s	MSE _y
(0.1, 0.9)	0.043 $\pm$ 0.002	0.015 $\pm$ 0.001	0.035 $\pm$ 0.035	0.009 $\pm$ 0.005
(0.3, 0.7)	0.042 $\pm$ 0.001	0.015 $\pm$ 0.001	0.045 $\pm$ 0.049	0.021 $\pm$ 0.036
(0.5, 0.5)	0.044 $\pm$ 0.002	0.016 $\pm$ 0.001	0.050 $\pm$ 0.046	0.021 $\pm$ 0.046
(0.7, 0.3)	0.042 $\pm$ 0.002	0.017 $\pm$ 0.001	0.042 $\pm$ 0.040	0.022 $\pm$ 0.033
(0.9, 0.1)	0.043 $\pm$ 0.002	0.018 $\pm$ 0.002	0.049 $\pm$ 0.030	0.018 $\pm$ 0.016
Ours	0.041$\pm$0.001	0.015$\pm$0.001	0.012$\pm$0.014	0.005$\pm$0.006

diction, and long-term outcome prediction. Specifically, β and γ represent the primary trade-off between the two outcomes; thus, we enforce $\beta + \gamma = 1$ for simplicity. To provide diverse initializations, we pre-train the model using the following (β, γ) pairs: $\{(0.1, 0.9), (0.3, 0.7), (0.5, 0.5), (0.7, 0.3), (0.9, 0.1)\}$. We withhold 20% of the training data as a validation set to guide parameter selection. For each experiment with a distinct random seed, the configuration yielding the lowest $\text{MSE}_s + \text{MSE}_y$ on the validation set is selected.

The intensity of the MI loss is governed by α . Compared to β and γ , model performance is relatively insensitive to the initial value of α . Through a grid search over $\{0.001, 0.01, 0.1\}$, we fixed $\alpha = 0.001$ as it yielded the most consistent validation performance. The results for these initial configurations are reported in the first five rows of [Tables 8](#) and [9](#). Building upon these initializations, the Pareto-Optimal module further explores the locally continuous frontiers for model fine-tuning. The final parameters are determined by the optimal performance achieved during this exploration phase, which is reported as the last row in the two tables.

The initial weights can greatly influence the performance. In general, assigning higher weights to a specific objective can yield a more accurate prediction of this single task, but exacerbates the imbalance among multiple objectives. For example, a clear decreasing trend is demonstrated in MSE_s but an increasing trend in MSE_y on the IHDP dataset. However, being too focused on the balance among multiple tasks can also lead to a local optimal solution. Like the results of Jobs, Twins, and Simulation datasets, the optimal solution may correspond to a more aggressive weight assignment. Therefore, utilizing the Pareto principle enables the exploration of more balanced or even superior results.

When the model is sensitive to these parameters (IHDP, Jobs, and Simulation), our strategy is more robust to reasonable variations in these hyper-parameters. For example, performance varies greatly on the Jobs dataset due to different pairs of (β, γ) and random seeds. But in each trial, our method can adaptively and stably choose the optimal model parameters, achieving 16.20% ~ 54.23% improvement in MSE_s and 19.91% ~ 61.86% in MSE_y .

5.3.5. Sensitivity to smoothness

To further evaluate the robustness of our framework when the smoothness assumption is mildly violated, we conduct a sensitivity analysis by introducing structured noise into the potential outcome functions defined in [Eq. \(21\)](#). Specifically, a sinusoidal component with a high frequency ($k = 50$) is used to simulate rapid local oscillations:

$$S \leftarrow S + \delta \cdot \sin(k \cdot T), Y \leftarrow Y + \delta \cdot \sin(k \cdot S), \quad (23)$$

where $\delta \in \{0.01, 0.05, 0.1\}$ is the hyper-parameter to control the intensity. Note that $\delta = 0$ is the scenario reported in [Section 5.3.1](#). By varying the noise intensity δ , we report the corresponding results in [Table 10](#).

Although the estimation errors of all models inevitably increase with the intensity of high-frequency noise δ , our framework consistently outperforms all baselines across all scenarios and maintains a stable and low MSE. This resilience suggests that our representation learning, guided by the Pareto-Optimal module, effectively filters out local high-frequency oscillations and captures the underlying causal mechanisms rather than over-fitting to non-smooth artifacts.

Table 10
Estimation results when violating smoothness.

	$\delta = 0.01$		$\delta = 0.05$		$\delta = 0.1$	
	MSE _s	MSE _y	MSE _s	MSE _y	MSE _s	MSE _y
TarNet	0.287 $\pm$ 0.242	0.214 $\pm$ 0.031	0.297 $\pm$ 0.260	0.213 $\pm$ 0.032	0.302 $\pm$ 0.276	0.220 $\pm$ 0.034
CFR	0.219 $\pm$ 0.129	0.163 $\pm$ 0.104	0.232 $\pm$ 0.142	0.175 $\pm$ 0.110	0.244 $\pm$ 0.144	0.183 $\pm$ 0.112
DRNet	0.272 $\pm$ 0.375	0.498 $\pm$ 0.368	0.319 $\pm$ 0.403	0.518 $\pm$ 0.381	0.325 $\pm$ 0.403	0.556 $\pm$ 0.418
VCNet	0.236 $\pm$ 0.140	0.180 $\pm$ 0.107	0.247 $\pm$ 0.151	0.184 $\pm$ 0.111	0.252 $\pm$ 0.162	0.189 $\pm$ 0.107
CRNet	0.031 $\pm$ 0.010	0.039 $\pm$ 0.013	0.035 $\pm$ 0.012	0.043 $\pm$ 0.017	0.044 $\pm$ 0.016	0.050 $\pm$ 0.019
FG	0.049 $\pm$ 0.022	0.023 $\pm$ 0.011	0.057 $\pm$ 0.024	0.034 $\pm$ 0.012	0.061 $\pm$ 0.024	0.039 $\pm$ 0.014
Ours	0.018[*]$\pm$0.006	0.014[*]$\pm$0.004	0.021[*]$\pm$0.010	0.018[*]$\pm$0.007	0.027[*]$\pm$0.012	0.021[*]$\pm$0.007

* means statistically significant results (p-value $\leq$ 0.05).

Table 11
Running time (minutes) of estimation.

	Simulation	IHDP	Jobs	Twins	Crime	MIMIC-III
TarNet	3.171	0.200	0.503	0.761	0.099	0.746
CFR	3.561	0.205	0.507	0.864	0.097	0.824
DRNet	3.525	0.201	0.675	0.901	0.092	0.765
VCNet	2.988	0.223	0.472	0.833	0.126	0.819
CRNet	3.819	1.672	1.223	1.266	0.142	1.288
DPPL	–	–	–	–	0.052	0.139
FairGrad	5.952	4.137	1.718	0.754	0.577	0.903
Ours	5.172	0.287	0.909	1.516	0.121	1.469

5.3.6. Computational complexity

When conducting the experiment of estimation of S and Y , we also report the training time required for each method. The results are summarized in Table 11. Generally speaking, a larger sample size, higher feature dimensionality, and a more complex model architecture could require more time for training. From the perspective of datasets, Simulation, Twins, and MIMIC-III are relatively large in scale, and the time cost required is generally higher. In terms of model design, our proposed POE-POPL and the multi-task learning approach (FairGrad) require more time. This is because they involve exploration of the Pareto frontier during parameter updates. Such modules bring about a significant improvement in performance. Overall speaking, the time cost of our model is acceptable compared to all these baselines, and it is clearly worthwhile in terms of the trade-off between time costs and performance.

6. Conclusion

In this paper, we present a novel Pareto-efficient framework designed to navigate the complex trade-offs between short- and long-term outcomes in treatment optimization. To address the dual challenges of task-level conflicts and gradient interference, we introduce two synergistic components: Pareto-Optimal Estimation (POE) and Pareto-Optimal Policy Learning (POPL). By integrating a continuous Pareto optimization module, our framework explicitly reconciles competing objectives during representation learning and identifies the optimal treatment-response frontier across continuous dosage spaces. Extensive evaluations on various datasets (synthetic and real-world) demonstrate the effectiveness of our method. Furthermore, sensitivity analyses also validate its robustness to hyper-parameter initializations and resilience against assumption violations.

CRedit authorship contribution statement

Yingrong Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis;

Anpeng Wu: Writing – review & editing, Validation, Methodology, Investigation, Formal analysis; **Haoxuan Li:** Methodology, Investigation, Formal analysis; **Weiming Liu:** Writing – review & editing, Validation, Formal analysis; **Baohong Li:** Writing – review & editing; **Qiaowei Miao:** Software; **Ruoxuan Xiong:** Writing – review & editing; **Fei Wu:** Writing – review & editing; **Kun Kuang:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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